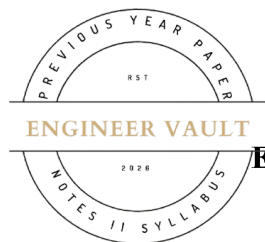


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**SUBJECT CODE NO: - RNP_8027****FACULTY OF SCIENCE AND TECHNOLOGY****F.E (Group B) (NEP) Sem - II****Examination November / December 2025 (To be held in February-2026)****Basics of Electronics Engineering****[Time: 3 Hours]****[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1

Solve the following questions**20**

a) Which component allows current in one direction only?

- a. Resistor
- b. Capacitor
- c. Inductor
- d. Diode

b) FET is a _____ controlled device.

- a. Current
- b. Voltage
- c. Power
- d. Capacitance

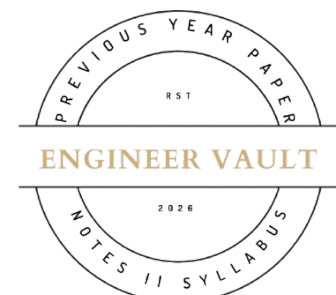
c) Which of the following is an active component?

- a. Resistor
- b. Capacitor
- c. Inductor
- d. Transistor

d) An op-amp as a comparator compares:

- a. two currents
- b. two voltages
- c. two resistors
- d. two capacitors

- e) Which of these is a common op-amp IC?
- a. IC 555
 - b. IC 7805
 - c. IC 741
 - d. IC 4017
- f) Which gate gives low output when both inputs are high?
- a. AND
 - b. XOR
 - c. NOR
 - d. NAND
- g) Full adder adds:
- a. 1 bit
 - b. 2 bits
 - c. 3 bits
 - d. 4 bits
- h) Which transducer is used for displacement measurement?
- a. Thermistor
 - b. Strain Gauge
 - c. LVDT
 - d. RTD
- i) Transducers that do not require an external power source are:
- a. Active
 - b. Passive
 - c. Both A and B
 - d. None
- j) Which of the following is not a transmission medium?
- a. Coaxial cable
 - b. Optical fiber



- c. Air
- d. Generator

SECTION B

- Q.2 a) Draw and Explain basic block diagram of DC Regulated power supply. **05**
- b) Explain 3 terminal fixed regulators in detail **05**
- Q.3 a) Write short note on Field Effect Transistor **05**
- b) What are the different configurations of BJT explain in detail. **05**
- Q.4. a) Explain different types of Number systems in detail with one example each. **05**
- b) Explain Op-Amp as a differentiator and Integrator **05**
- Q.5. a) Explain different types of Number systems in detail with one example each. **05**
- b) Classify and Explain different types of logic gates. **05**
- Q.6 a) Write short note on LVDT. **05**
- b) Explain any two temperature measurement transducers **05**
- Q.7 a) Draw and Explain the block diagram of different elements of communication system **05**
- b) State and explain different transmission media **05**

